

THE ESTIMATOR DECIDES: WHAT CURRICULA AND FAILURE RECYCLING CAN AND CANNOT DO IN RL WITH VERIFIABLE REWARDS

Anonymous authors

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ABSTRACT

Post-training with verifiable rewards increasingly relies on two data-level interventions: *difficulty curricula*, which reallocate rollout budget toward learnable prompts, and *failure recycling* (hindsight relabeling), which converts failed rollouts into verified successes of the tasks they actually achieved. Both are treated as objective-agnostic add-ons. We show they are not: the advantage estimator underneath decides whether each helps or harms, in closed form. For success-conditioned (MaxRL-style) advantages, the expected advantage mass a prompt emits from N rollouts is exactly $2(\text{pass}@N - \text{pass}@1)$ — a compute-indexed learnability functional whose band location, width, and scaling follow from the rollout budget alone, and whose two zeros partition difficulty into a dead zone no sampler can reach and a mastered tail where updates buy sharpening instead of signal. Curricula can only reallocate compute within the band between them; recycling is the only channel into the dead zone; and relabels that land in the mastered tail sharpen. This predicts an estimator-conditioned divergence we then measure: under identical curricula, every success-conditioned run grows $\text{pass}@k$ coverage and every GRPO run loses it (nine maze runs, exact permutation $p=0.0079$; the teacher-deficit direction replicates 2/2 seeds at LLM scale). It also predicts a failure mode of recycling we then observe in production — hindsight arms improve mean accuracy while losing coverage, **recycling-induced sharpening** — and supplies the mitigation: gating relabel admission by the destination’s pass rate converts the loss into a tunable mean-versus-coverage dial. The boundary of the safety claim is measured too: a pre-registered fifth suite whose learnable band collapses to one template stratum defeats *every* estimator — the partition needs a graded band to act on. One diagnosis covers all of it: audit the estimator before shipping a curriculum or a recycler, and measure coverage, the currency in which every failure mode here is visible.

1 INTRODUCTION

Post-training language models and control policies with verifiable rewards has a cost structure unlike supervised learning: the data is free, but *rollouts* are not. Every optimization step is preceded by sampling N completions per task, generation dominates wall-clock (85% of step time, averaged over 272 logged steps across our GSM8K sessions; telemetry committed), and on hard task pools most of that compute buys nothing — under uniform sampling we measure that **65–75% of rollout groups produce zero learning signal**: either every rollout fails (the group is dropped by success-conditioned estimators) or every rollout succeeds (no contrast, no gradient).

The field’s response is a pair of data-level interventions. The first is the *difficulty curriculum*: reallocate the rollout budget toward tasks where learning is possible right now — UCB bandits over difficulty buckets (Wang et al., 2025), target-difficulty controllers (Shi et al., 2025), category bandits (Chen et al., 2025), learnability-based rejection sampling (Foster et al., 2025), dead-prompt filtering (Yu et al., 2025; Zheng et al., 2025). The second, newer, is *failure recycling*: convert a failed rollout into a verified success of the task it actually achieved — hindsight relabeling, recently

brought into RLVR-style loops for robot policies (Xu et al., 2026). Both interventions are, in current practice, designed and evaluated as if they were *objective-agnostic*: bolt-on modules whose benefit is assumed to transfer across the advantage estimator they sit on, and whose cost is assumed visible in the mean-accuracy metrics they are tuned against.

This paper shows that neither assumption survives contact with the estimator’s own algebra. The organizing fact is a partition. **The estimator’s expected advantage mass $u_N(p)$ divides task difficulty into a dead zone, a learnable band, and a mastered tail. Curricula can only reallocate compute within the band; recycling is the only channel into the dead zone; and relabels that land in the mastered tail buy sharpening, not signal. The estimator decides all three.** Three questions unpack this:

Q1: What can a curriculum contribute, at most? For success-conditioned advantages, the expected advantage mass a task emits from a group of N rollouts has a closed form: $2(\text{pass}@N - \text{pass}@1)$, a compute-indexed learnability functional whose two zeros are the partition’s boundaries (§3). Mass is a surrogate for learning signal (Remark 1), but it is the part a sampler can act on before spending a rollout, and it makes the allocation *derivable* rather than heuristic: the band’s location, width, and N -scaling come from the budget you already chose, and published learnability curricula are the $N=2$ slice. It also bounds the channel. Allocation can only redistribute signal that exists: a perfect-information allocator ties our cheap posterior in end performance (§6.1), and at LLM scale a prompt-level teacher starves before it pays (§6).

Q2: When is either intervention safe? The same algebra says the answer depends on the estimator. MaxRL concentrates $(N-1) \times$ more mass than RLOO on hard tasks as $p \rightarrow 0$ and releases mastered tasks; GRPO’s variance-normalized weights keep spending on both tails. We predict and measure the resulting divergence at two scales: on the maze, every MaxRL run grows pass@8 coverage and every GRPO run loses it (§6.3); on GSM8K, the pre-registered prediction that GRPO+teacher would be the only regressing cell landed on the registered run (§6.7). Recycling has its own failure mode, and the partition predicts it: relabels that land in the mastered tail sharpen. On Countdown arithmetic this is **recycling-induced sharpening** — exact-verifier relabeling improves mean accuracy while *losing* pass@ k coverage — the data-induced sibling of GRPO’s weight-induced collapse, formally hindsight’s marginal-distribution shift (Ghosh et al., 2021) composed with curation-driven self-consumption (Ferbach et al., 2024).

Q3: Does the algebra also supply the fix? Yes — by applying the functional a third time. A relabel to a destination the model already reaches sits in the mastered tail, where mass is zero, so we gate relabel admission by the destination’s estimated pass rate: the same utility that schedules sampling, now deciding what the recycler may admit. Across three seeds the gate restores frontier coverage to the no-recycling baseline while keeping most of recycling’s mean gain, and sweeping its strength traces a monotone mean-versus-coverage trade — a dial, not a switch (§6.9). Where recycling is healthy the gate is transparent: hard-destination relabels pass untouched, so the validated testbeds reproduce their results with the gate enabled. One method, no per-domain switches.

Our contributions:

1. **A closed-form curriculum (§3):** the expected advantage mass of a prompt under success-conditioned RL is exactly $2(\text{pass}@N - \text{pass}@1)$, a compute-indexed learnability functional whose band, width, and N -scaling are derived rather than tuned, and whose zeros partition difficulty into the dead zone, the band, and the mastered tail; published learnability curricula are its $N=2$ slice.
2. **A derived teacher and a characterized recycler (§4–5):** FrontierMax schedules prompts by this functional with the per-group estimator untouched, and hindsight relabeling under a verified-success contract yields the ML gradient of the achieved task under the original sampling law.
3. **Estimator-conditioned predictions, confirmed (§6):** under identical curricula, success-conditioned runs grow pass@ k coverage while GRPO runs lose it, across four task suites from exact-gradient chains to LLM post-training.
4. **A new failure mode and its derived fix (§6.8–6.9):** exact-verifier recycling improves mean accuracy while losing coverage — recycling-induced sharpening — and gating relabels by destination pass rate converts the loss into a monotone mean-versus-coverage dial.

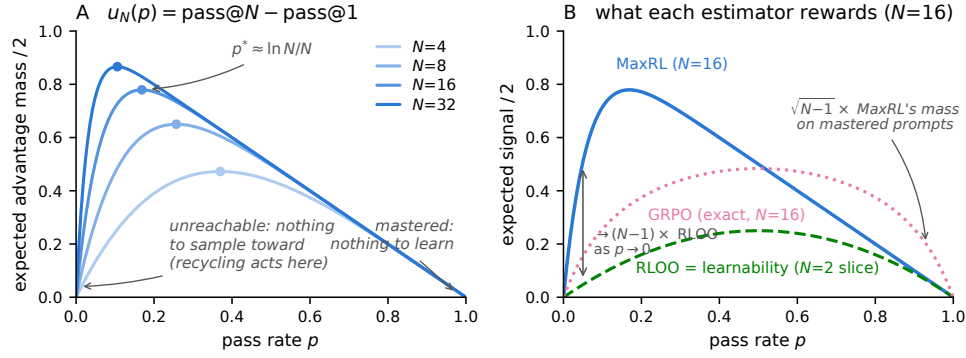


Figure 1: **The estimator is the curriculum.** *Left:* the derived utility $u_N(p) = (1 - (1 - p)^N) - p$ — the estimator’s expected advantage mass (Prop. 1) — for growing rollout budgets N . The peak $p^* \approx \ln N/N$ walks toward harder prompts as compute grows; both tails are dead zones the weight function cannot reach. *Right:* all three curves are exact expected mass at $N=16$ (GRPO’s is $\frac{1}{N}\mathbb{E}\sqrt{K(N-K)}$, $K \sim \text{Bin}(N, p)$; MC-verified). MaxRL concentrates $\rightarrow (N-1) \times$ more mass than RLOO on frontier prompts (Prop. 2) and releases mastered prompts fastest; GRPO’s variance normalization keeps $\sim \sqrt{N-1} \times$ MaxRL’s relative mass on mastered prompts — the maintenance-of-easy-tasks its curriculum arms lose in §6. RLOO’s profile is exactly the published “learnability” objective — the $N=2$ slice.

Along the way we state a refinement to the MaxRL paper — the practical drop- $K=0$ estimator is unbiased for the $T=N-1$ objective, not $T=N$ (Lemma 1) — and every prediction below was committed before the deciding runs, with negatives and retractions reported in place (§8) and audited in the repository.

2 BACKGROUND: MAXRL IN THREE LINES

For binary rewards, maximum likelihood maximizes $\mathbb{E}[\log p_\theta(\text{success} \mid x)]$. MaxRL expands $\log p$ as a Maclaurin series in $(1 - p)$ and truncates at order T , giving a compute-indexed family $J^{(T)}$ interpolating REINFORCE ($T=1$) to exact ML ($T \rightarrow \infty$); the practical estimator sets per-rollout advantages $w_i = r_i/K - 1/N$ with all-fail groups dropped (Lemma 1 gives its exact truncation order). Its weight function grows as $p \rightarrow 0$: the objective is an *implicit, gradient-level* curriculum. Our work is the data-level complement: the same algebra, read as a sampling rule plus a recycling rule.

3 THE ESTIMATOR IS THE CURRICULUM

Lemma 1 (Truncation order of the practical estimator). *The drop- $K=0$ variant of the practical MaxRL estimator is exactly unbiased for the $T=N-1$ member of the objective family, not $T=N$: zeroing the control variate only on all-fail groups makes it outcome-dependent, contributing $-(1 - p)^{N-1}\nabla p$, and $\sum_{k=1}^N (1 - p)^{k-1} - (1 - p)^{N-1} = \sum_{k=1}^{N-1} (1 - p)^{k-1}$.*

Interpretation. A refinement to [Tajwar et al. \(2026\)](#), immaterial for their conclusions but load-bearing for our mass accounting at small N — all identities below are for the estimator as actually shipped.

Proposition 1 (Advantage mass). *For a prompt with pass rate p and N i.i.d. rollouts under the practical MaxRL weights,*

$$\mathbb{E}\left[\sum_i |w_i|\right] = 2(\text{pass}@N(p) - \text{pass}@1(p)) = 2((1 - (1 - p)^N) - p) \equiv u_N(p). \quad (1)$$

Proof (three lines): on $\{K \geq 1\}$, $\sum_i |w_i| = K(\frac{1}{K} - \frac{1}{N}) + \frac{N-K}{N} = 2(1 - \frac{K}{N})$, and $\mathbb{E}[2(1 - \frac{K}{N})\mathbf{1}\{K \geq 1\}] = 2(\Pr[K \geq 1] - p) = 2(\text{pass}@N - \text{pass}@1)$.

Interpretation. The estimator’s expected learning signal on a prompt is twice the probability that the prompt is solvable within N attempts but not within one. That sentence is a curriculum, and its two zeros are a **partition of difficulty**: a dead zone ($p \rightarrow 0$: nothing to condition on), a learnable band peaking at $p^* = 1 - N^{-1/(N-1)} \approx \ln N/N$, and a mastered tail ($p \rightarrow 1$: no contrast left). The band’s center, width, and compute-scaling are set by the one parameter you already chose. Everything in this paper is one of three moves against this partition: allocate within the band (§4), create signal in the dead zone (§5), and refuse updates that land in the mastered tail (the gate).

Proposition 2 (Frontier amplification). As $p \rightarrow 0$, the ratio of MaxRL’s expected signal to RLOO’s ($2p(1-p)$ exactly) tends to $N-1$.

Interpretation. The comparison to GRPO is two-sided and exact: GRPO’s expected mass is $\frac{1}{N} \mathbb{E} \sqrt{K(N-K)}$ with $K \sim \text{Bin}(N, p)$, and taking tails, $\text{MaxRL/GRPO} \rightarrow \sqrt{N-1}$ as $p \rightarrow 0$ while $\text{GRPO/MaxRL} \rightarrow \sqrt{N-1}$ as $p \rightarrow 1$. MaxRL over-serves the frontier and releases mastered prompts; GRPO silently keeps $\sqrt{N-1} \times$ more of its mass on mastered prompts — maintenance a curriculum then removes. Concentrating sampling on the frontier only helps if the estimator extracts signal there, and §6 shows the mastered-tail difference becoming an active failure under a curriculum.

Proposition 3 (Allocation). The rollout allocation maximizing total expected advantage mass is greedy water-filling on the marginal $p(1-p)^N$ — the probability that the next rollout is a group’s first success.

Interpretation. This is the mass-optimal reference allocator, not the deployed sampler: the practical teacher (§4) keeps fixed group sizes and samples prompts $\propto u^\gamma$, trading optimality for group-parallel batching and for the exploration a point-optimal allocator forfeits (the unconstrained maximizer concentrates on the current arg-max, a brittle choice under posterior noise). We measure what the gap costs: a true-pass-rate water-filling oracle ties the Thompson teacher in end performance (§6.1).

Remark 1 (Scope of the surrogate). Advantage mass counts weighted rollouts, not score-vector geometry. We use it as the sampling utility because (i) it is knowable *before* spending rollouts, (ii) its zeros are exact — at $p \in \{0, 1\}$ the estimator emits no gradient of any norm — and (iii) on the exact-gradient testbed it *ties* the true first-order expected improvement as a task ranking (Spearman within .005 at every snapshot; artifact `results_bridge.json`). What it does not claim to be is the optimal sequential utility, and no state-local functional is: on independent tasks the true final-performance optimum is a planning solution (trajectory-knapsack with a budget-dependent funding cutoff), and which surrogate best approximates it depends on the training objective — a variance-tilted $(1-p)^{\alpha} u_N$ wins anytime performance while model-predictive replanning wins the final checkpoint, each losing the other’s metric (8–10/10 paired seeds each way; App. A). “Best curriculum utility” is ill-posed until that objective is fixed; we report both currencies throughout. The partition’s boundaries — the load-bearing claim — are the zeros every candidate shares.

Remark 2 (The curriculum changes the training distribution). Sampling prompts from an adaptive q_t instead of the data distribution ρ optimizes a reweighted objective $\mathbb{E}_{x \sim q_t} [J(x)]$: each group’s gradient is an unbiased MaxRL gradient *for its own prompt*, but the mixture is tilted toward the frontier, without importance correction back to ρ . This is the standard bargain every curriculum strikes (uniform sampling makes the same choice for $q_t = \rho$), and we treat it as an empirical question, not a free lunch: all evaluations in §6 are on fixed held-out pools under ρ , so any mismatch cost is charged to the method. The safety results are precisely about when this tilt helps or hurts the ρ -measured outcome.

One identity, three literatures: learnability curricula (Rutherford et al., 2024; Foster et al., 2025) are the $N=2$ slice of u_N ; DAPO-style dynamic sampling and GRESO are its avoidance shadow (cull where $u \approx 0$); HER-style relabeling (Andrychowicz et al., 2017) is its creation complement (manufacture $K > 0$ where $u = 0$ identically; §5). DisCO (Li et al., 2025) and SC-SDPO (Liu et al., 2026) derive analogous per-question weights for the GRPO family to diagnose difficulty bias; our identity generalizes that line to success-conditioned estimators and — the step neither takes — uses it to predict how *external* curricula and recyclers interact with each estimator, in coverage terms.

The partition, as a map of the paper (Fig. 3). The **teacher** (§4) allocates compute within the band — and saturates: a true-pass-rate oracle allocator ties the cheap posterior in end performance (§6.1), so better pass-rate information is not where remaining gains live. **Recycling** (§5) creates signal in the dead zone — the only channel that still adds on top of the oracle. The **estimator** underneath

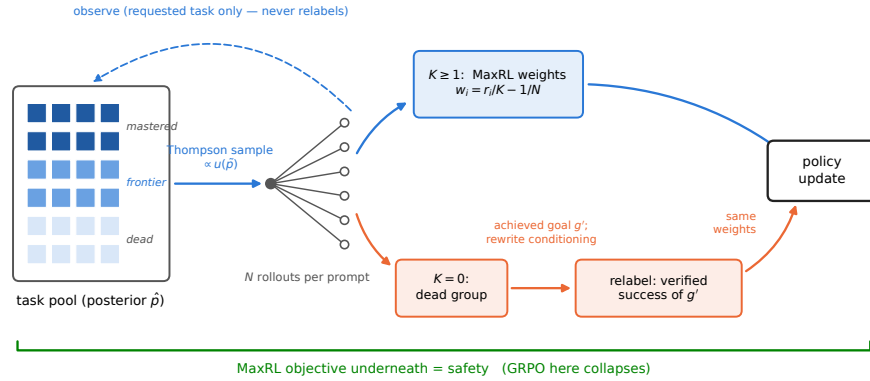


Figure 2: **One FrontierMax step.** A Thompson-sampled posterior allocates rollouts by the derived utility; live groups train with unmodified MaxRL weights; dead groups are relabeled to the sub-goal they actually achieved (conditioning rewritten) and train as verified successes. The posterior observes requested-task outcomes only.

Algorithm 1 FrontierMax: one step. Three lines differ from a standard group-rollout loop (marked $\triangleright$).

- 1: $\triangleright$ **sample** prompts $x_1..x_B \propto u_N(\tilde{p}_x)^\gamma$ mixed with a uniform floor, $\tilde{p}_x \sim \text{Beta}(\alpha_x, \beta_x)$ (else: uniform)
- 2: roll out N completions per prompt; verify; $K_x \leftarrow$ successes
- 3: update posterior (α_x, β_x) from (K_x, N) for the requested task only
- 4: live groups ($1 \leq K_x$): MaxRL advantages $w_i = r_i/K - 1/N$, unchanged
- 5: $\triangleright$ **dead groups** ($K_x=0$): relabel to the achieved sub-goal g' (verified; conditioning rewritten)
- 6: $\triangleright$ admit the relabel only if $\hat{p}_{g'} \leq \text{gate_max_p}$ (the gate: refuse the mastered tail)
- 7: one policy-gradient step on live + admitted relabeled groups

decides whether either is safe, and its mastered tail is where both failure modes live (§6). One line: *the teacher allocates, hindsight creates, the estimator decides whether either is safe.*

4 FRONTIERMAX: THE SCHEDULE

FrontierMax is a standard group-rollout RLVR loop with three modified lines (Alg. 1). A decayed Beta posterior tracks each prompt’s pass rate from observed group outcomes only — never from relabeled successes, which inflate it ($\hat{p}$ 0.81 vs. eval 0.47). Thompson sampling draws $\tilde{p}$ and samples prompts $\propto u(\tilde{p})^\gamma$ with a uniform floor. The knobs (decay 0.7, floor 0.1, $\gamma \in \{1, 4\}$) carry validated defaults (Table 3); what is *derived* rather than tuned is the difficulty band itself — location, width, and N -scaling — which is exactly where competing curricula spend their hyperparameters.

5 FAILURE RECYCLING

Proposition 4 (Hindsight update, characterized). *Relabeling a dead group to the sub-goal g' its trajectories actually achieved, rewriting the goal conditioning, and applying the same success-conditioned weights yields the ML gradient of task g' under the conditional law induced by the original task’s sampling; it equals the fresh-rollout gradient exactly when the two conditional laws match, and is otherwise a distribution-shifted (biased) estimate of it.*

Interpretation. “Exactness” is a property to measure per environment, not a blanket guarantee. On our validation testbed the laws match by construction, so cosine probes there are consistency checks, not evidence (several null relabeling schemes produce identical cosines); the evidence that the relabel direction carries information is behavioral — the placebo decomposition below. What the distribution shift costs in general is an open, environment-specific question.

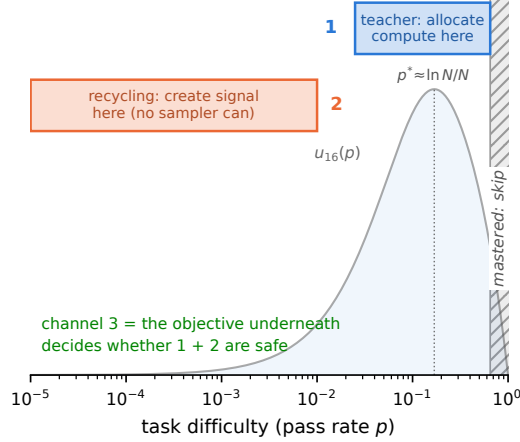


Figure 3: **The partition as a map of the method.** The teacher allocates compute within the learnable band; recycling creates signal in the dead zone no sampler can reach; the gate refuses relabels that land in the mastered tail; the estimator underneath decides whether any of it is safe.

Two contracts keep practice on the proof’s side: **(1) exactness** — a relabeled success must be a true success of the relabeled task under the environment’s own verifier (never an LLM judge); **(2) conditioning** — goal-embedding trajectories must have their conditioning rewritten to the achieved goal. Skipping (2) turns the exact gradient-target into noise: on our gridworld, hindsight without the rewrite scores below teacher-only ($0.600 < 0.658$); with it, it leads (0.703). Recycling is the only mechanism here that *creates* signal, and we decompose its headline gain rather than let it inflate: of the $+0.22$ AUC on fixed task pools, a placebo replaying *live* gradients on dead-group slots captures 83% — most of the benefit is extra gradient dose on otherwise-wasted slots — while the relabel *direction* carries $+0.037$ beyond the strongest placebo, is the tightest arm in the battery, and is exactness-dependent (random-target relabels lose to dose-matched replay 5/5). Both numbers ship together. The value is regime-dependent in a way we can state precisely: $+0.22$ AUC on fixed task pools, $+0.01$ on one-shot task streams — the gain is proportional to how much a relabeled skill can compound, and task-set fixedness is the single most important regime variable we found.

6 EXPERIMENTS: AN ESCALATING LADDER

Each rung asks one question of the partition, at increasing scale and decreasing experimental control; Table 1 is the map. Protocol, stated once: $\pm$ denotes standard deviation across seeds; paired contrasts share per-seed warmstarts; permutation tests are exact (all label assignments); “coverage” is pass@ k at the k named per suite (8 on the maze, 16 on Countdown, 4 on GSM8K); AUC is the mean of the eval curve over training at matched wall-clock or episode budget. Pre-registered predictions were committed before the deciding runs; negative results appear in place, one sentence each, with full postmortems in the repository’s audit trail.

6.1 Does allocation saturate? (exact-gradient skill chains; CPU, 5 seeds). Uniform $0.650 \rightarrow$ teacher $0.728 \rightarrow$ full stack 0.890 ± 0.002 AUC. An eight-arm control battery (App. A) orders into a *gradient-information ladder*, each rung a 5/5 paired-seed win: zero-information dose ($0.798\text{--}0.800$) $<$ replayed correct direction (0.832) $<$ true relabels (0.869) $<$ perfect allocation (oracle $0.8885 \approx$ full stack 0.8895) $<$ allocation+creation (0.8935). Two decompositions matter downstream. *Allocation saturates*: a floor- and γ -matched true-pass-rate oracle ties the full stack — perfect difficulty information is worth almost nothing over the cheap posterior. *Creation decomposes*: a replay placebo captures 83% of recycling’s $+0.22$; the relabel *direction* carries $+0.037$ beyond it and is the most reliable arm in the battery, while random-target relabels lose to dose-matched replay 5/5. The utility form is load-bearing through its N : the N -blind learnability slice $p(1-p)$ forfeits the entire allocation gain ($-.007$ vs. uniform) where u_N carries $+.078$ (5/5). All arms converge within .027 of one another — on fixed pools these interventions buy *time-to-ceiling*, not the ceiling.

Table 1: The ladder at a glance: every rung asks one question of the partition; scale rises, control falls.

rung	question	setting	seeds	outcome
6.1	does allocation saturate?	skill chain (exact grads)	5	yes: oracle ties posterior
6.2	anything to sample toward?	frontier-heavy pool	5	no: only creation scores
6.3	main effect, real gradients?	maze transformer, 1.26M	9 runs	yes: 9/9, $p=0.0079$
6.4	where/when does coverage go?	maze, level-resolved	22 runs	easy band, immediately
6.5	when does allocation not pay?	IsaacLab Anymal-C	1	learnable-everywhere: null
6.6	beyond task spread?	MountainCar/CartPole	3	spread=existence; teacher=speed
6.7	LLM scale?	GSM8K 2×2, 360M	1–2	sign landed; replication open
6.8	what does recycling cost?	Countdown v2	3	mean up, coverage down
6.9	does the gate recover it?	Countdown v2	3	yes: a monotone dial

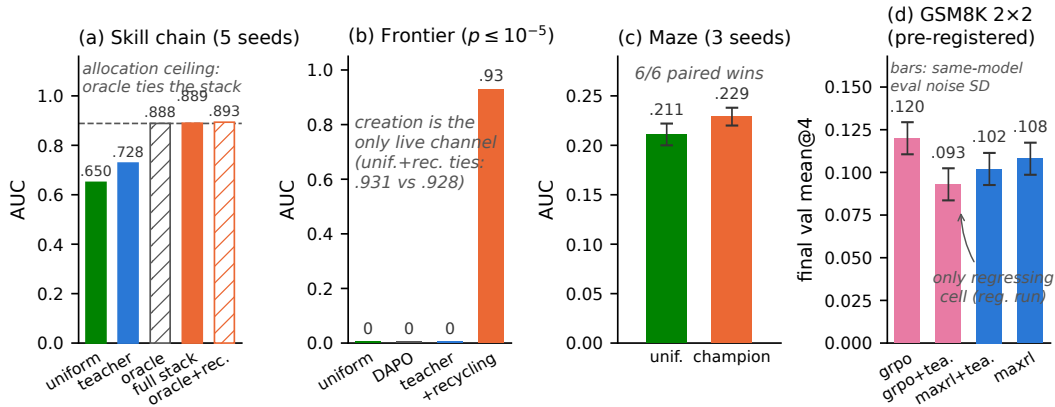


Figure 4: **The ladder.** (a) Exact-gradient chains: a floor-and- γ -matched true-pass-rate oracle *ties* the full stack — allocation saturates — and recycling still adds on top of the oracle. (b) Frontier-heavy regime: allocation scores exactly zero; recycling solves it, and does so equally under uniform sampling (.931 vs. .928) — creation, not allocation, is the live channel. (c) Maze at matched wall-clock: 6/6 paired seed wins. (d) GSM8K 2×2, registered run, error bars = measured same-model eval noise (SD .0094): GRPO+teacher is the only cell that regresses (pre-registered P-G2; a replication seed did not reproduce it — §6.7); the teacher’s MaxRL effect is a null at this budget (P-G1).

Takeaway. A monotone information ladder: dose < direction < allocation-ceiling < +creation. Allocation saturates; creation adds on top; every intervention buys speed, not asymptote.

6.2 Can any sampler act when nothing is learnable? (frontier-heavy regime, max pool pass rate 10^{-5} ; 5 seeds, artifact committed). Uniform, DAPO, and the plain teacher score **exactly 0.00**; adding recycling reaches **0.98 final (0.93 AUC)** — relabeling invents the curriculum below the pool, ignites within ~ 400 groups, then goes silent. The attribution control pins down which channel did it: *uniform+recycling* scores the same ($0.931 \pm .002$ vs. *teacher+recycling* $0.928 \pm .003$ AUC) — in this regime the entire effect is signal creation; allocation contributes nothing because there is nothing learnable to allocate toward. Equivalently: a designer who hand-added the achieved sub-goals to the pool would also escape the zero — what recycling buys is *discovering* that sub-pool from the policy’s own failures, with no task engineering.

Takeaway. Where nothing is learnable at budget, sampling policy is irrelevant — uniform and teacher tie at zero without recycling and tie at 0.93 with it. Creation is the only live channel, and its value is discovery, not allocation.

6.3 Does the estimator main effect survive real gradients? (maze transformer at matched wall-clock; 1.26M params, 13 levels, ~ 30 runs, 3 seeds). Champion (teacher + dense recycling)

0.252±0.005 final / 0.229±0.009 AUC vs. uniform 0.230±0.015 / 0.211±0.011; paired deltas positive 6/6 seeds; 22–35% more optimization steps per GPU-hour. Attribution within that gain, stated plainly: matched *wall-clock* lets curriculum arms take 1.2–1.4× more optimization steps (dead groups skip the backward pass), and step-matched truncation shows the extra steps are most of the pure teacher’s effect (step-matched teacher-only $\Delta\text{AUC} +0.006$, $p\approx0.36$, n.s.) while teacher+recycling survives step-matching ($p\approx0.04$) — on the maze the teacher’s measured value is *throughput* (buying more steps per GPU-hour), recycling’s is signal. The maze teacher uses the legacy frontier form; a pre-registered 3-seed rerun with the exact u_N measures the two within noise *on the maze itself* (AUC .2138±.0059 vs. .2135±.0104; artifact `un_uniform_verdicts.json`), closing the gap between the deployed form and the derived one. The same sweep refutes a CPU-derived within-band refinement (a $(1-p)$ -tilted utility that won 10/10 on the testbed loses 0/3 on the maze on every meter) — within-band utility choices are testbed-sensitive, which is why the theory claims only the band’s boundaries. Safety, stated as what the design supports: an estimator *main effect on coverage* — all five MaxRL runs grow pass@8 (e.g. 0.316 → 0.348 under the teacher) and all four GRPO runs lose it (0.308 → 0.271 uniform; 0.332 → 0.269 with the teacher), permutation $p=0.0079$ on the 9-run main effect. Whether the teacher *amplifies* GRPO’s decay is a factorial (interaction) question this suite is not powered for; the GSM8K 2×2 below tests exactly that cell, pre-registered. Inference currency (one convention, every level: smallest k reaching absolute coverage 0.25; single checkpoint pair): our checkpoint needs 11× fewer samples at level 5, ties GRPO at levels 2–3, and needs 3× more at level 4 — the frontier is where the efficiency gain lives. The summed-coverage view agrees without a target choice: GRPO ahead at pass@1, curves crossing at $k\approx4$, MaxRL ahead by half a level at $k=64$ (App. A) — choose the estimator by how many samples you will draw at inference.

6.4 Where and when does GRPO’s coverage go? (level-resolved). Decomposing the main effect by difficulty band turns it into a mechanism (Fig. 5). *Where:* GRPO’s pass@8 loss is concentrated almost entirely on the easy/mid band L1–3 (−.224 mean over its four runs, every run negative) while its L1–3 pass@1 *rises* (+.02 to +.14, every run) — it funds reliability by liquidating easy-band coverage; MaxRL holds L1–3 flat (+.005±.058 over 18 runs) and concentrates its gain on the frontier band L4–6 (+.126±.063 vs. GRPO’s +.057). The band asymmetry is exact-permutation $p=0.0001$. This is the observable consequence of the mass curves in Fig. 1: the $\sqrt{K(N-K)}$ weighting keeps spending updates on mastered prompts (sharpening them), where u_N has already released them. *When:* the divergence is immediate, not saturation-triggered — GRPO incurs 59–73% of its total diversity-premium loss (pass@8 − pass@1: .123 → .042) within the first quarter of the budget, with the first easy-band pass@8 drop visible at steps 0–50 when L1 pass@1 is still only .62–.75; MaxRL’s premium stabilizes after warmup (.122 → .101). A dose-response note from the same logs: sparse relabeling (1/dead group) uniquely preserves the diversity premium (exact $p=0.007/0.008$ vs. none/dense) while dense banks the same coverage into the best final pass@1 — dose sets *where* the coverage dividend is spent, not whether coverage is lost.

Takeaway. The gains survive real gradients, and the coverage main effect now has a located mechanism: GRPO liquidates easy-band coverage for reliability from the first updates; MaxRL reallocates the same budget to the frontier. Coverage is the currency that sees it.

6.5 When does allocation *not* pay? (IsaacLab legged locomotion, pre-registered null). A 5-arm Anymal-C rough-terrain pilot (one seed, run by an independent team on a co-designed protocol; App. A) lands its pre-registered null exactly: on a terrain grid where every level is learnable at budget, the stock greedy curriculum beats the frontier teacher outright (mean pass .372 vs. .278). With the maze step-matched analysis and Countdown, the regime rule is a three-scale result: **allocation pays only where unlearnable-at-budget regions exist to avoid** — on learnable-everywhere pools, breadth beats focus.

6.6 What does a curriculum add beyond task spread? (Gymnasium control — binary-success RLVR on classic control; official-task currency). All arms train on *binary task success* (MountainCar: reach the flag; CartPole: survive 400) — the classic-control analogue of a verifier reward. The **target-only** arm is then a provable zero-update freeze, not a slow learner: every group is all-fail, every group-relative estimator emits exactly zero weights on constant rewards, and telemetry confirms zero parameter updates over the full budget (random-policy success $< 3\times10^{-6}$; no budget ignites it). The gated full stack under the identical reward reaches **1.000±0.000 on both official tasks** (3

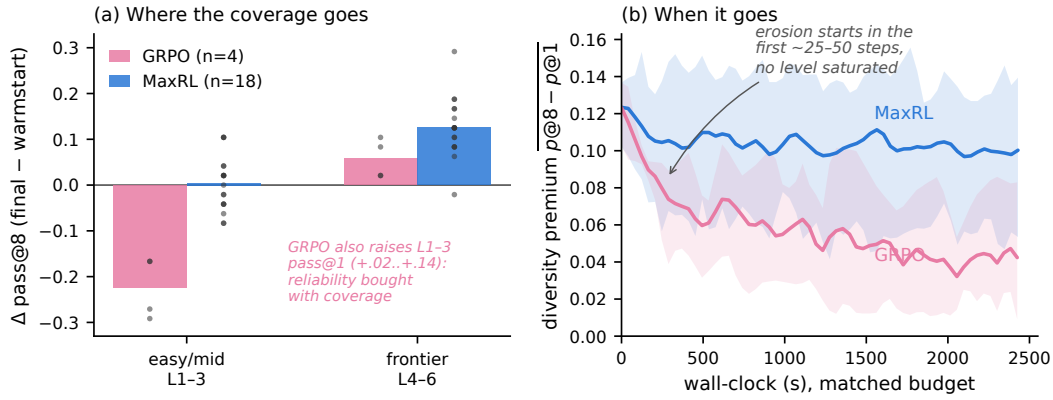


Figure 5: **Where and when GRPO’s coverage goes (maze, level-resolved, matched wall-clock).** (a) $\Delta \text{pass@8}$ (final – warmstart) by difficulty band: GRPO’s loss lives on the easy/mid band (all 4 runs; dots are runs) while its pass@1 there rises — sharpening; MaxRL holds the easy band and gains on the frontier band (18 runs; band asymmetry exact permutation $p=0.0001$). (b) Diversity premium $\text{pass@8} - \text{pass@1}$ vs. wall-clock (mean and min–max across runs): GRPO’s erosion begins within the first ~ 25 –50 steps, before any level saturates; MaxRL stabilizes after warmup.

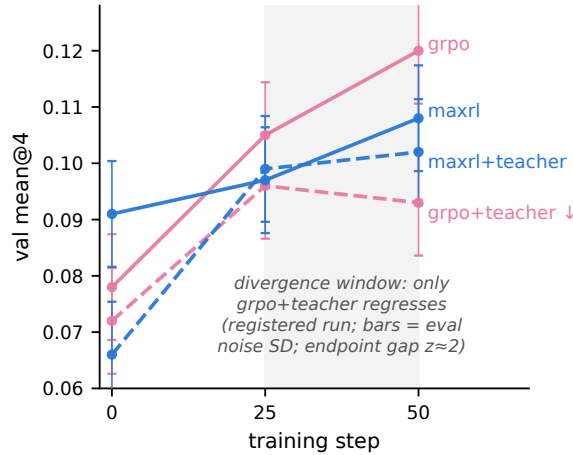


Figure 6: **GSM8K 2×2 (pre-registered; registered run shown; error bars = measured same-model eval noise, SD .0094).** GRPO+teacher is the only cell that regresses in its second half (P-G2 on this run; the load-bearing statistic is the endpoint gap vs. plain GRPO, $z \approx 2$ — the second-half dip alone is within noise; a replication seed with weaker measured steering climbed instead — \$6.7); under MaxRL the identical teacher trains stably to its best value but does not beat uniform at this budget (P-G1: null).

seeds, Fig. 8). A dense-reward control confirms the zero is a property of sparse verifier-style reward (REINFORCE on CartPole’s native reward solves it; App. A). Attribution at a common horizon: uniform-over-bins task spread supplies most of the effect (existence of a gradient), recycling-without-allocation plateaus below ceiling, and the teacher’s remaining contribution is speed and stability ($3 \times$ faster to hard-task 0.9, 2.5 – $6 \times$ smaller cross-seed variance). A methods note we keep because it generalizes: our own first version used per-bin task-partitioned policy parameters and reproduced the known failure (0.000 in every arm) — curricula operate through shared parameters, or not at all.

Takeaway. Curricula operate through shared parameters, or not at all.

6.7 Does it transfer to LLM scale? (GSM8K 2×2, pre-registered; SmolLM2-360M, N=16, one A10G). Predictions committed before any cell finished. **P-G2 landed on the registered run:** GRPO+teacher was the only cell that regressed in its second half, and the deficit against plain GRPO at the endpoint is 2–4× the measured same-model eval noise (mean@4 $-.027$, $z = -2.0$; pass@4 $-.048$, $z = -2.0$; the second-half dip itself, $.096 \rightarrow .093$, is within the $\pm .009$ noise floor and carries no weight on its own) while plain GRPO climbed monotonically ($.078 \rightarrow .105 \rightarrow .120$). Steering telemetry, read against its own controls: the teacher’s sampled-dead fraction *minimum* reaches .48 vs. uniform’s .52, but the run means are indistinguishable (.66–.67 in all cells) — the delivered treatment was weak, which is the posterior-starvation mechanism below, and it bounds how much of the GRPO gap the curriculum can own. **A replication seed does not reproduce the regression shape:** seed 2 of the same cell climbs monotonically ($.073 \rightarrow .095 \rightarrow .118$), and its steering telemetry shows the treatment was weaker (min dead-sampled fraction .53, mean $.66 \approx$ population — a near-uniform teacher on average). The completed *within-seed* pair (grpo-uniform seed 2: $.081 \rightarrow .085 \rightarrow .125$, final pass@4 .229; artifact `grpo_uniform_seed2.json`) sharpens the read: the endpoint teacher deficit under GRPO has the same *sign* in both seeds on both meters (4/4 signed contrasts; seed 2: $-.007$ mean@4, within noise, $-.026$ pass@4, $\sim 1.5\times$ its floor) with magnitude tracking steering intensity — a dose effect, not a sign flip. The pre-registered *second-half-regression* shape remains 1 of 2 seeds: suggestive, not established. MaxRL+teacher trains stably to its best value ($.066 \rightarrow .102$); the uniform-MaxRL cell finishes at .108, making P-G1 (a teacher gain for MaxRL) a *null at this budget* — predicted in advance by the posterior-starvation analysis below. A pass@ k sweep on the registered checkpoints: the teacher’s deficit under GRPO *widens monotonically with k* ($-.008$ at $k=1$ to $-.024$ at $k=16$) — curriculum damage lives in coverage, as the maze predicted, though each individual delta is within single-seed noise (harness caveats in App. A). One design confound is disclosed rather than dismissed: the teacher samples with replacement, so curriculum cells revisit prompts more — a repetition effect is an alternative reading that a uniform-with-replacement control cell separates; it is running with the steering-controlled replication. The mechanism behind the weak treatment: the teacher’s posterior *learns* real difficulty from ~ 1 visit per prompt ($\rho = -0.17$ against independent 7B-model solve-rate annotations, $p \approx 10^{-17}$; cross-run posterior agreement $3\times$ its agreement with the external key — it learns *this model’s* landscape) but its allocation is *posterior-starved*: 3,200 draws over 7,473 prompts leave the estimated utility landscape nearly flat, so Thompson sampling stays near-uniform (full posterior diagnostics in App. A). Small model; the pre-registered outcomes were orderings and signs on the registered run.

Takeaway. The maze main effect stands on nine runs; the LLM-scale interaction is 1-of-2-seeds and tracks steering intensity — the decisive experiment is a steering-controlled multi-seed cell, queued.

6.8 What does recycling cost? (Countdown 2×2, pre-registered): two nulls and a new phenomenon. The first exact-verifier hindsight experiment in RLVR ran in production (12 relabeled groups/step, 72–89% yield). Both positive predictions returned null, with an exact diagnosis: the uniform baseline reached 94/76/94% of the random-guesser bound implied by the pool’s simplified construction — no headroom for allocation or creation to matter. What the run found instead is new: **recycling-induced sharpening** — hindsight arms improved mean@16 while *losing* pass@16 coverage (tier-1: .571 vs. .645 for the no-relabel baseline). Recycled off-target successes concentrate the policy on reachable outputs at the expense of exploration breadth: the creation channel has its own safety trade-off, mirroring §6.3’s weight-induced collapse — data-induced rather than weight-induced, and to our knowledge unreported (no prior work measures pass@ k under hindsight in RLVR). Our algebra predicts it: a relabel to a self-achieved value is a task at $p \approx 1$, where $u(p)$ says training signal is zero — the softmax pays for those updates out of the exploration tail. *Takeaway: recycling needs an admission rule, and the estimator already provides it — the same derived utility that schedules sampling should gate the relabel destination.*

6.9 Does the gate recover it? (Countdown v2 pool, 3 seeds, pre-registered). We rebuilt the pool with real structure (random operand permutation, parenthesization, exact division; a headroom gate excluded guesser-saturated tiers) and ran three arms $\times$ three seeds: no recycling (B1), ungated recycling (B2), gated recycling (B3). Setup note with a theory reading: recycling has a *syntactic prerequisite* — the pre-SFT base model produces zero verifier-valid failures (relabel yield 0.000, $n=384$), so there is nothing to recycle until SFT installs the output format; post-SFT the model is a near-independent guesser whose pass@16 sits at a tier-invariant $\sim 75\%$ of the independence

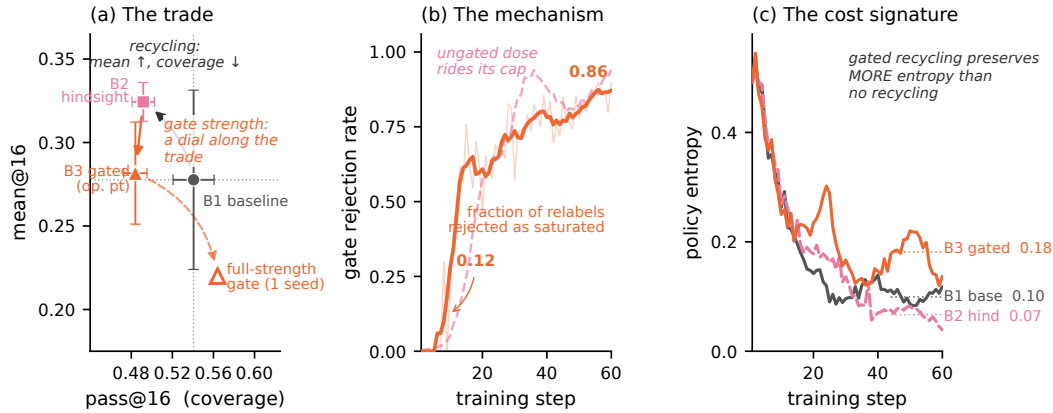


Figure 7: **Recycling-induced sharpening and its derived mitigation (Countdown, tier 1).** (a) Three seeds, both axes: recycling (B2) buys mean@16 with pass@16 coverage. On this tier the moderate gate (B3) gives back most of the mean without moving coverage — tier-1 destinations saturate early, so admitted relabels are few — while full-strength gating (corrected decay, 1 seed) returns coverage *above* baseline at the mean’s full price: gate strength is a dial along the trade, and its useful middle lives on the frontier tier, where the same moderate setting restores coverage to baseline (.279±.019 vs. .274) keeping ~60% of the mean gain (\$6.8). (b) The mechanism in-run: the gate’s rejection rate rises monotonically as the model’s reachable set saturates its relabel destinations, while the ungated dose rides its cap. (c) Gated recycling preserves *more* policy entropy than no recycling — admitted relabels are a spread-preserving gradient.

bound $1 - (1 - \text{pass}@1)^{16}$ (ratios .744/.783/.755 across tiers) — the baseline our headroom gate measures against. **Sharpening replicates:** at tier 1, recycling buys mean@16 +.046 (.278±.054 → .324±.012) and pays -.049 of pass@16 (.541±.020 → .492±.011), both outside seed noise; the frontier tier moves the same direction. **The timing is dose-then-damage:** recycling’s frontier-tier mean gain is installed by step 30 — +.014±.008 paired, all seeds positive, at only ~28% of the eventual relabel dose — while the coverage cost only materializes by step 45 (−.048±.019 paired, *all seeds negative*, after ~65% of dose): the benefit precedes the damage, so early-stopping on mean-accuracy validation would bank the gain and never see the bill. The damage is also visible on the training side: the ungated arm ends with more unsolvable-at-16 frontier training prompts than baseline in every seed; the gate’s entropy retention replicates across seeds while ungated recycling makes entropy seed-unstable (details in App. A). **The gate works where the algebra says it should:** at the frontier tier (destinations not yet saturated) the gated arm returns coverage to baseline (.279±.019 vs. .274±.015) while keeping ~60% of the mean gain; at tier 1 (destinations saturate early, so the gate blocks nearly all relabels) it neutralizes recycling almost entirely. Implementation disclosure: a decay-math bug made all three-seed gated runs gate *weaker* than designed (effective destination threshold $\hat{p} \approx 0.64$ where the design implies 0.92), so the reported operating point is an under-gated one; the corrected-decay rerun (1 seed) gates ~3× harder and pushes tier-1 coverage *above* baseline at the mean’s full price (Fig. 7a). Together the three arms trace a monotone dose-response — no gate: full mean gain, coverage lost; under-gated: ~60% of the gain, coverage restored; full-strength: gain erased, coverage above baseline — `gate_max_p` is a dial along a mean-vs-coverage frontier, and the bug, favorably for the method, sampled its middle. In-run telemetry shows the mechanism (Fig. 7b-c): rejection rises .12 → .85 as the reachable set saturates, and the gated arm ends with *more* policy entropy than the no-recycling baseline. A pre-registered `gate_max_p` dose sweep at fixed decay is the standing follow-up.

Takeaway. Sharpening is a reproducible property of exact-verifier recycling, and the derived gate converts it into a tunable mean-vs-coverage trade whose useful range sits where destinations are not yet saturated.

7 RELATED WORK

The coverage tension. RLVR itself trades coverage for precision: $\text{pass}@1$ rises while $\text{pass}@k$ at large k falls below the base model (Yue et al., 2025), with entropy collapse as the mechanistic signature (Cui et al., 2025) and overtraining on saturated problems as a locus (Yuan et al., 2026). Our contribution to this line is the *interaction*: whether external curricula and recyclers compound or cancel the estimator’s implicit weighting, and what that costs in coverage. Notably, recent work frames curricula as the *cure* for coverage loss (Cai et al., 2026); our GRPO arms show the cure is estimator-conditional — those studies never varied the estimator. The estimator–difficulty interaction itself is a heating 2026 theme — SC-SDPO (Liu et al., 2026) shows GRPO’s normalization implicitly encodes a difficulty weighting, and A-GRAE (Yu et al., 2026) redesigns the advantage around difficulty focus — but no prior work runs *identical* curricula under two estimators or measures the resulting $\text{pass}@k$ sign flip.

Curricula for RLVR. Difficulty-adaptive sampling is bucket- or scalar-level and heuristic-scored: DUMP (UCB over buckets), AdaRFT (scalar controller over precomputed labels, TMLR 2026), SEC (category bandit), threshold curricula. LILO (NeurIPS 2025) is the closest per-prompt method — rejection sampling by $\hat{p}(1 - \hat{p})$, which Prop. 1’s corollary identifies as the $N=2$ slice of our derived utility. Reinforce-Ada (Xiong et al., 2025) — the adaptive sampler MaxRL itself points to — adapts the rollout count *within* a prompt until a signal criterion is met, holding the prompt distribution fixed; our teacher is the complementary move, reallocating *across* prompts at fixed group size, and Prop. 3 is the idealized form of both (water-filling spends its next rollout wherever $p(1 - p)^N$ is largest, across or within). DAPO’s dynamic sampling and GRESO cull dead prompts — avoidance without allocation or creation. PKPO and Pass@k-training modify the objective toward $\text{pass}@k$ but keep uniform sampling. **Hindsight and self-training for LLMs.** Exact-verifier relabeling has precedent outside RLVR loops — CodeIt (Butt et al., 2024) (ARC program synthesis, relabel + prioritized replay) and proof-tree relabeling (Poesia et al., 2024) — and LLM-judged variants relabel agent trajectories (AgentHER, HSL). Concurrent 2026 work brings relabeling-adjacent mechanisms into RLVR-style training: LfH (Xu et al., 2026) (a VLM proposes one shared hindsight instruction per failed group for VLA post-training; the relabeler is a judge, their stated limitation) and ZipRL’s hindsight response replay (Hu et al., 2026) (densifying GRPO signal for multi-turn agents). None of these measures $\text{pass}@k$ under relabeling — precisely the currency where we find recycling’s hidden cost. The nearest phenomenon report is in same-problem rejection finetuning: Strozzi (2026) document a STaR-trained model winning $\text{pass}@8$ while the base model overtakes at $\text{pass}@256$ — concentration over reach — for SFT on a model’s own verified outputs. Recycling-induced sharpening is the cross-task, on-policy-RL sibling, and our gate is its estimator-derived admission rule; gating *vanilla* RLVR updates by saturation signals has independent precedent (Yuan et al., 2026’s Bayesian boundary gating), not applied to relabel destinations. So the precise claims we retain: no prior work (i) derives the sampling utility from the estimator’s own expected signal, (ii) couples it with success-conditioned weights it provably matches, or (iii) *measures the coverage cost of hindsight relabeling in RLVR* and gates relabel admission by destination saturation.

8 LIMITATIONS AND NEGATIVE RESULTS

Scale. The LLM-scale teacher×estimator interaction is 1-of-2 seeds and tracks steering intensity: the replication seed of the GRPO+teacher cell did not reproduce the registered regression, and its telemetry shows the treatment was barely delivered (§6.7). The GSM8K teacher is also coverage-neutral-to-negative under *both* estimators at this budget, so the contrast there concerns the sign of the mean-accuracy effect, not a coverage rescue. The maze estimator main effect (nine runs, zero exceptions) is the safety result this paper stands on; a steering-controlled multi-seed LLM cell is the decisive next experiment.

Scope of the surrogate. Advantage mass is not gradient norm, SNR, or expected improvement (Remark 1), and §6.5 exhibits the regime where reallocating it buys nothing (learnable-everywhere pools). Recycled-gradient exactness is a property of the environment’s relabel map; noisier verifiers will land between our fixed-pool and one-shot endpoints (+0.22 vs. +0.01 AUC).

The coverage main effect is pool-conditional. A pre-registered fifth suite (Jugs water-measuring puzzles; postmortem and artifacts in the repository) returned all-null: on a pool whose learnable band is a single stratum of 1–2-move template solutions, *plain MaxRL* also collapses — entropy $1.36 \rightarrow 0.07$ – 0.21 in 60 steps, frontier-tier pass@16 $0.15 \rightarrow 0$, every arm converging to the same deterministic policy — and recycling accelerates the collapse rather than rescuing it. When N rollouts of a near-trivial plan are near-identical, the i.i.d.-rollout diversity the group contrast requires is absent, and success-conditioned weights liquidate frontier coverage through shared parameters just as GRPO’s do elsewhere. The estimator main effect of §6.3–6.4 is therefore a claim about pools with a graded band at the deployed N , not an unconditional safety guarantee; per-task rollout-set diversity at initialization is the design gate we now check first. The same suite also exposed a granularity bug in the gate (keying destinations by value collides tasks when achieved values are few and shared — ≤ 20 integers in Jugs vs. hundreds in Countdown); the corrected gate keys on the relabeled task.

What did not transfer. A $4\times$ duration run refuted our own duration hypothesis (the deep-frontier stall is a per-step-competence ceiling; depth needs capacity, not schedule); γ -concentration did not transfer from chains to the maze, as the compounding model predicted; adaptive truncation order underperformed fixed T ; and feeding relabeled successes to the teacher’s posterior inflates it, so the posterior sees requested-task evidence only.

Corrections along the way. Two early claims were retracted on internal review — a teacher-steering read traced to an epoch-frozen sampler (the fixed run confirmed the effect), and a “beats the oracle” comparison traced to a floor handicap in the oracle arm (§6.1 reports the corrected tie). Full postmortems live in the repository’s audit trail.

REFERENCES

- Tajwar, F., Zeng, G., et al. Maximum Likelihood Reinforcement Learning. *ICML*, 2026.
- Li, G., Lin, M., Galanti, T., Tu, Z., Yang, T. DisCO: reinforcing large reasoning models with discriminative constrained optimization. *NeurIPS*, 2025.
- Liu, Z., Cao, Y., Chen, J., Honavar, V. G. Restoring the sweet spot: pass-rate weighted self-distillation for LLM reasoning. arXiv:2605.27765, 2026.
- Yue, Y., et al. Does RL really incentivize reasoning capacity in LLMs beyond the base model? *NeurIPS* (oral), 2025.
- Yuan, S., et al. Understanding diversity collapse in RLVR via the lens of overtraining. arXiv:2606.15455, 2026.
- Cui, G., et al. The entropy mechanism of RL for reasoning language models. arXiv:2505.22617, 2025.
- Cai, W., et al. Curriculum RL can incentivize reasoning capacity beyond the base model. arXiv:2606.22317, 2026.
- Wang, Z., Cui, G., Li, Y.-J., Wan, K., Zhao, W. DUMP: automated distribution-level curriculum learning for RL-based LLM post-training. arXiv:2504.09710, 2025.
- Shi, T., Wu, Y., Song, L., Zhou, T., Zhao, J. Efficient reinforcement finetuning via adaptive curriculum learning. *TMLR*, 2026. arXiv:2504.05520.
- Chen, X., et al. Self-evolving curriculum for LLM reasoning. arXiv:2505.14970, 2025.
- Foster, T., Sims, A., Forkel, J., Foerster, J. LILO: learning to reason at the frontier of learnability. *NeurIPS*, 2025.
- Yu, Q., et al. DAPO: an open-source LLM reinforcement learning system at scale. *NeurIPS*, 2025.
- Xiong, W., et al. Reinforce-Ada: an adaptive sampling framework under non-linear RL objectives. arXiv:2510.04996, 2025.
- Zheng, H., et al. Act only when it pays: efficient reinforcement learning for LLM reasoning via selective rollouts (GRESO). *NeurIPS*, 2025.
- Andrychowicz, M., et al. Hindsight experience replay. *NeurIPS*, 2017.
- Ghosh, D., et al. Learning to reach goals via iterated supervised learning. *ICLR*, 2021.

- Ferbach, D., et al. Self-consuming generative models with curated data provably optimize human preferences. *NeurIPS*, 2024.
- Xu, I., Jiang, S., et al. Learning more from less: reinforcement learning from hindsight. arXiv:2607.09042, 2026.
- Butt, N., et al. CodeIt: self-improving language models with prioritized hindsight replay. *ICML*, 2024.
- Poesia, G., et al. Learning formal mathematics from intrinsic motivation. *NeurIPS*, 2024.
- Hu, Z., Wang, L., Wang, X., et al. ZipRL: adaptive multi-turn context compression with hindsight response replay. arXiv:2605.28069, 2026.
- Strozzi, I. L. Teacher-free self-training amplifies but does not compound: a pass@ K crossover on a free-verifier domain. arXiv:2606.07856, 2026.
- Yu, Z., Chen, Z., Liu, M., et al. Unveiling implicit advantage symmetry: why GRPO struggles with exploration and difficulty adaptation. arXiv:2602.05548, 2026.
- Rutherford, A., et al. No regrets: investigating and improving regret approximations for curriculum discovery. *NeurIPS*, 2024.

A FULL DETAILS FOR COMPRESSED EXPERIMENT RUNGS

Material moved from §6 during the page-budget pass; every number here is artifact-backed in the repository.

6.1 control battery (full). Eight arms, 5 seeds each: uniform 0.650, lr $\times$ 2 0.798, random-target relabels 0.800, live-gradient replay 0.832, true relabels 0.869, no-floor γ -matched oracle 0.8885 $\pm$.0014, full stack 0.8895 $\pm$.0023, oracle+recycling 0.8935 $\pm$.0012. The relabel direction term (+.037 over replay) has seed spread $\pm$.0025, 4–6 $\times$ tighter than every placebo; its worst seed beats every placebo’s best. Random-target relabels are harmful relative to dose-matched replay (−.032, 5/5 seeds) but neutral vs. plain lr-doubling: fake labels waste the slot rather than poison training. Utility-form controls: exact u_N 0.728 $\pm$.002 vs. legacy frontier form 0.733 $\pm$.013 (the derived form’s payoff is 7 $\times$ smaller seed variance); the $N=2$ slice $p(1-p)$ is a wash against uniform (−.007, 2/5 wins) where u_N carries +.078 (5/5). AUC spread across arms is 9 $\times$ the final-performance spread.

6.5 IsaacLab protocol. Five arms: frontier teacher, stock greedy walker, uniform, scripted ramp, no-motion control; fixed-grid per-level evaluation; Anymal-C rough terrain; one seed, executed by an independent team against a co-designed, pre-registered protocol. The teacher’s only edge over the greedy walker is a within-noise sliver at the hardest evaluated level.

6.6 gym control (full). MountainCar flag: measured random-policy success $< 3\times 10^{-6}$ over 10^6 episodes; survive-400 $\approx 10^{-13}$ by tail fit. Dense-reward controls (prediction registered before the run): CartPole’s native +1/step reward lets vanilla REINFORCE on the same tile policy solve survive-400 (0.96 $\pm$ 0.03, 3 seeds); MountainCar’s native −1/step reward (constant until first success) stays at 0.000 in all seeds — the categorical zero tracks reward sparsity, not the environment. Attribution at a common 280-step horizon (hard-task AUC): MC .750 $\rightarrow$ +recycling .895 $\rightarrow$ +teacher .956; CartPole .708 $\rightarrow$.792 $\rightarrow$.893; paired teacher increment positive 3/3 seeds on both; creation-without-allocation plateaus at 0.958 on CartPole in every seed while adding the teacher restores 1.000; teacher speed effect 3.0 $\times$ to hard-task 0.9 (2.1 $\times$ to 0.99 on CartPole). The per-bin ablation replicates the 10-seed transition-matched study (flag 0.848 $\pm$ 0.058) before strengthening it to full convergence.

6.3 inference-efficiency detail. Convention: smallest k reaching absolute coverage 0.25, log-interpolated; artifact maze_gpu/efficiency.json; Fig. 9 plots the summed-coverage crossing. Summed coverage over L0–6: GRPO 3.24 vs. MaxRL 3.07 at pass@1; crossing at $k\approx 4$; 5.12 vs. 4.62 at $k=64$, with GRPO’s level-2/4 curves saturating at 0.88/0.56 where MaxRL’s reach 1.00/0.62.

6.9 secondary reads (full). Training-side damage: ungated recycling ends with more unsolvable-at-16 frontier training prompts than baseline in every seed (paired +.083/+.063/+.034). Entropy at 3 seeds: the gate’s retention replicates (+.072 paired over baseline, all seeds positive, $\sim 1.6\times$); ungated recycling’s endpoint entropy is seed-unstable (spread $\pm$.062, 4–9 $\times$ the other arms’), consistent with its noisy frontier-coverage endpoints.

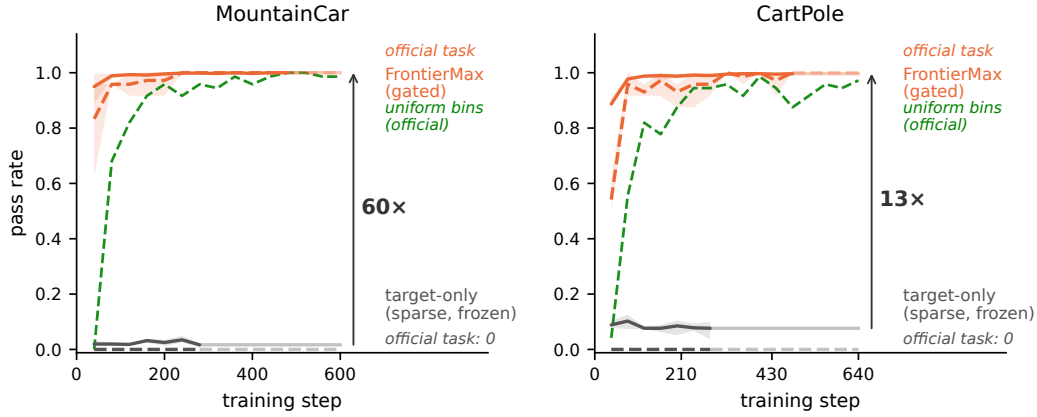


Figure 8: **Gym control under binary task-success reward, trained to plateau (3 seeds).** Solid: mean pass rate across curriculum bins; dashed: the *official* task (MountainCar flag / CartPole survive-400). The target-only arm (gray) is a provable zero-update freeze under sparse verifier-style reward; a uniform-over-bins control (green) also cracks the task, $3\times$ slower and off-ceiling; the gated FrontierMax stack converges to 1.000 ± 0.000 .

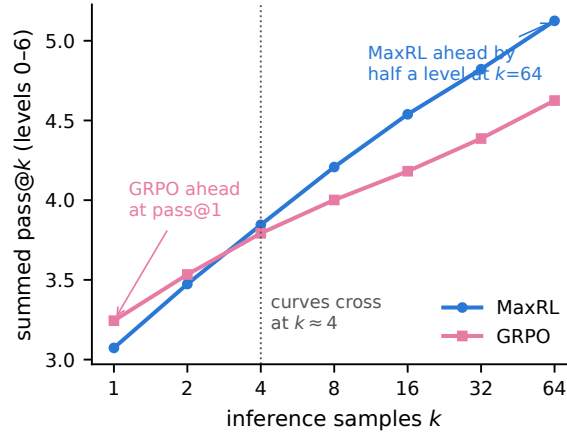


Figure 9: **Summed maze coverage vs. inference samples k (final uniform-arm checkpoints).** GRPO leads at pass@1; the curves cross at $k \approx 4$; MaxRL leads by half a level at $k=64$ — choose the estimator by how many samples you will draw at inference.

6.7 posterior diagnostics and harness caveats (full). The pass@ k sweep’s absolute level sits $\sim 3\times$ below the trainer’s val metric (unreconciled harness discrepancy; only the within-sweep contrast is quoted). Posterior detail: two independent teacher runs’ posteriors agree with each other (cross-run Spearman $\approx .45$ on co-visited prompts, $n \approx 900$; $\approx .67$ self-consistency with a shared latent) vs. $.10-.22$ against external 7B-model annotations, from ~ 1.25 visits per prompt. Across external-difficulty quintiles spanning solve rates $.33 \rightarrow .98$, mean $\hat{p}$ moves only $.076 \rightarrow .103$ — the compressed landscape that starves allocation. Response length separates runs by estimator, perfectly: all three GRPO runs compress to 209.6 ± 1.8 tokens while both MaxRL runs sit at 232.4 ± 5.7 (common ~ 365 -token warmstart; the teacher moves length ≤ 3 tokens) — sharpening visible in a currency where token entropy is ambiguous. Prompt-uniqueness figures for the replacement confound: $\sim 34\%$ unique prompts seen by teacher arms vs. $\sim 43\%$ uniform.

Jugs suite (pool-conditionality; §8). Tier grid t0–t4 (2–5 jugs, disjoint min-moves bands, 200 unique tasks/tier); feasibility landscape (SmoLLM2-360M two-shot, $N=16$): t0 .058/.475 (pass@1/pass@16), t1 .006/.100, t2–t4 0/0 with relabel-yield-on-failure .62–.73 on every tier. All

nine pre-registered B-arm cells (no-recycling / ungated / gated $\times 3$ seeds) converged to the same deterministic fixed point: $t0 \text{ mean@16} = \text{pass@16} = 0.450$ exactly (the 2-move template stratum), $t1 \text{ pass@16 } 0.15 \rightarrow 0$, actor entropy $1.36 \rightarrow 0.02\text{--}0.21$. Ungated recycling ran 169 relabels/step at yield .88 and ended with the lowest entropy (0.02–0.03); the value-keyed gate rejected 99.8% of relabels (the granularity bug described in §8, since fixed and regression-tested). Postmortem and per-cell artifacts in the repository.

B HYPERPARAMETERS AND COMPUTE

Total compute for every experiment in this paper: ~ 230 A10G-hours on a single GPU plus CPU-days for the exact-gradient suites; generation dominates LLM step time (85%, artifact `generation_timing.json`). Table 2 gives the per-suite configuration and Table 3 the knob inventory with provenance.

Table 2: Per-suite configuration. Teacher knobs (decay, floor, γ) are shared across suites with the defaults below; what varies per suite is scale, not schedule.

	chains (§6.1–6.2)	maze (§6.3–6.4)	GSM8K (§6.7)	Countdown (§6.8–6.9)
policy	exact tabular	1.26M transformer	SmolLM2-360M	SmolLM2-360M (SFT)
rollouts N	16	32	16	16
groups/step	1	8	64	64
optimizer, lr	SGD, 0.5	AdamW, 10^{-4}	AdamW, 10^{-5}	AdamW, 10^{-5}
budget	3,200 groups	2,400 s clock	50 steps	60 steps
seeds (headline)	5–10	3	1–2/cell	3/arm
compute/run	CPU-min	0.7 A10G-h	~ 20 A10G-h	~ 6 A10G-h

Table 3: Teacher and recycler knobs: value, provenance, and measured sensitivity.

knob	default	derived or tuned	sensitivity
band location/width/ N -scaling	—	derived (Prop. 1)	the theory claim
posterior decay	0.7	tuned (V2b)	closes $\sim 19\%$ of oracle gap
uniform floor	0.1	tuned	0.15 on maze; robust 0.05–0.2
γ (concentration)	1	tuned (V6)	4 on chains; no transfer to maze
gate_max_p	0.5	derived zero at $p \rightarrow 1$, midpoint	tuned monotone dial (§6.9)
relabel dose cap	12 groups/step	tuned	dose sets <i>where</i> coverage is spent (§6.4)

REPRODUCIBILITY

Every proposition has a Monte-Carlo verification script; every figure in this paper regenerates from JSON artifacts committed to the repository (`paper/figures/`, with LLM-run endpoints vendored in `paper/figures/data/`); the documentation passed a two-round adversarial audit. Known gaps, stated rather than papered over: the IsaacLab pilot is reported from an independent team’s summary (raw logs live with that team), and per-step timing fractions come from verl’s in-run telemetry in the ray session logs. The framework is a dependency-light package (`frontier_rl`, numpy-only core) with adapters for LLM pools (verl), gym control, IsaacLab parallel sim, and flow-policy VLAs.